

The register-mean polar bias as a symmetry witness: error mitigation, constrained optimization and qubit characterization in qg units

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Draft of 26 September 2026; revised with the trapped-ion and syndrome sections

Abstract

In the qg framework [1] a qubit is described by its polar bias $qg_Z = \langle \sigma_z \rangle$. When a circuit conserves a Hamming weight, the average of qg_Z over the register is fixed in advance: $\overline{qg}_Z = 1 - 2N/n$ for N excitations on n qubits. This one number is a reference-free witness of noise that changes the conserved quantity, and its shot-level version is the known symmetry-verification filter [2, 3]. We measure when the witness and the filter help, and when they do not, on six problems with classical controls, using calibration-based and vendor noise models. (i) On H_2 the filter lowers the energy error from 18 to 5.5 mHa, below Hartree–Fock (20.3 mHa), and it stays below Hartree–Fock along the dissociation curve. (ii) Across noise channels the witness predicts the outcome: it moves under T1 and the filter removes 94–95% of the error for free, while under dephasing it stays at its ideal value and the filter does nothing. Zero-noise extrapolation is complementary and fails on idle decoherence; the combination reaches 2.0 ± 3.9 mHa. (iii) With separate spin-up and spin-down witnesses, a second filter pays off only when a spin-leak witness is nonzero. Qubit routing on a heavy-hex device creates such leak in 1D Hubbard dynamics. (iv) In QAOA with an “exactly K ” constraint, the filter doubles the success probability of the constraint-preserving ansatz. Penalty QAOA at depth ≤ 2 does not beat a random feasible guess, and a greedy algorithm beats all quantum variants. (v) The witness relies on calibrated readout. A heralded sweep separates thermal population from asymmetric readout error, which the standard calibration conflates, and gives an unbiased effective temperature (± 1 mK) together with T_1 and T_2 . (vi) On IonQ’s trapped-ion noise models the filter doubles the constraint-preserving QAOA success and halves the H_2 error in every run, while zero-noise extrapolation, which must fold native gates because CX folding is undone by the compiler, is nearly unbiased but too noisy at the 2000-shot limit. (vii) Beyond symmetries, every stabilizer expectation of the repetition and Leung codes is a power of one single-qubit qg, so the syndromes track noise drift for free and choose between codes by the rule $T_2 > T_1$. We also report where the method stops: a 6-qubit LiH circuit and Hubbard circuits beyond ~ 190 two-qubit gates, where the witness correctly reports unital scrambling. No result is a quantum advantage over classical computation. Every number is reproduced by a script and pinned by a regression test.

1 Introduction

Near-term quantum computations are limited by noise, and error mitigation trades extra measurements or post-processing for lower bias [4–6]. When the target state has a known symmetry, such as particle number in a fermionic simulation or a cardinality constraint in optimization, outcomes that break it can be discarded. This is symmetry verification [2, 3].

This paper reads that symmetry in the language of the qg framework [1], where a qubit is described by $qg_Z = \cos \theta = \langle \sigma_z \rangle$. The contribution is not the filter, which is known, but three things around it:

1. a single, reference-free number, the register mean $\overline{qg}_Z$, that tells *before* any mitigation whether noise is breaking the symmetry and in which direction;
2. a systematic comparison, with classical baselines and a standard mitigation method (zero-noise extrapolation), of when the filter helps and when it cannot;
3. a calibration protocol for the readout on which the witness depends, written in the same units.

We state limits alongside results. In particular, every problem studied is classically easy at the sizes used, so the comparisons rank strategies for a quantum computation and do not claim a quantum advantage.

2 The witness

Proposition 1. *Let $W = \sum_i (1 - Z_i)/2$ count the qubits in $|1\rangle$. For any state supported on the eigenspace $W = N$,*

$$\overline{qg}_Z \equiv \frac{1}{n} \sum_{i=1}^n \text{Tr}(\rho Z_i) = 1 - \frac{2N}{n}. \quad (1)$$

Proof. $\frac{1}{n} \sum_i Z_i = 1 - \frac{2}{n}W$, and $W = N$ on the support of ρ . □

In the Jordan–Wigner encoding W is the electron number. In a “choose exactly K ” problem it is the cardinality. For a circuit that conserves W , the ideal $\overline{qg}_Z$ is therefore known for every value of the parameters. We use four quantities, all read from the same Z-basis shots:

- the *witness* $\overline{qg}_Z$. Amplitude damping (T1) moves it towards +1. Unital noise (depolarizing, dephasing) moves it towards the value of a maximally mixed register, 0;
- the *kept fraction*, i.e. the probability of the shots with $W = N$. It is 1 ideally and $\binom{n}{N}/2^n$ for a maximally mixed register, which marks the noise floor;
- the *filter*: keep only those shots and renormalize (symmetry verification);
- for two conserved numbers (spin up and spin down), one witness per register and a *spin leak*, the fraction of right- N shots in the wrong spin sector.

Observables measured in rotated bases, such as the $XXYY$ terms of a molecular Hamiltonian, cannot be filtered this way. The witness is linear in ρ , so its estimator from counts is an unbiased sample mean.

Readout. Readout errors enter as an affine map. If a qubit reads 1 when in $|0\rangle$ with probability e_{01} and 0 when in $|1\rangle$ with probability e_{10} , then

$$qg_Z^{\text{meas}} = a + b qg_Z, \quad a = e_{10} - e_{01}, \quad b = 1 - e_{01} - e_{10}. \quad (2)$$

An asymmetric readout ($a \neq 0$) therefore shifts the witness the way T1 does. Section 8 shows how to calibrate a and b without confusing them with the thermal population of the qubit.

Table 1: H_2 on `fake_brisbane`, energy error vs FCI (mHa), mean of 3 seeds. Hartree–Fock: 20.3 mHa.

idle delay	$\overline{qg}_Z$	kept	raw	+ readout	+ readout + qg filter
0 μs	+0.011	0.91	74	18	5.5
20 μs	+0.057	0.83	140	90	20
50 μs	+0.122	0.72	231	188	30

3 Methods

Noise models. (a) `fake_brisbane`: Qiskit Aer [7] with the noise model built from an IBM device’s calibration data (ECR gates, heavy-hex connectivity, thermal relaxation during scheduled idle times, readout errors), on a chain of the best-calibrated qubits. An optional idle delay before readout amplifies T1. (b) Controlled channels applied to every CX (so that gate folding scales them exactly): amplitude damping, phase damping or two-qubit depolarizing noise of strength p , or symmetric readout error only. (c) A generic all-to-all device dominated by depolarizing two-qubit error (0.6% per CX, 0.1% dephasing, 0.5% readout), with trapped-ion-like figures but not a vendor calibration. (d) IonQ’s `aria-1` and `forte-1` noise models, run on IonQ’s cloud simulator.

Mitigation. All quantum estimates include tensored readout-error mitigation unless stated. Zero-noise extrapolation (ZNE) [4, 5] folds every two-qubit gate G into G^3 and G^5 and extrapolates to zero noise with Richardson weights $(15, -10, 3)/8$. “ZNE + qg” filters at every noise scale and then extrapolates.

Statistics. Unless stated, 20,000 shots per circuit and mean $\pm$ standard deviation over 10 simulator seeds. The references are exact classical results (FCI, exact diagonalization, brute force) at these sizes.

4 Chemistry

H_2 (STO-3G, 0.735 Å) in the Jordan–Wigner encoding uses 4 qubits and 2 electrons, so the ideal $\overline{qg}_Z$ is 0. A number-conserving ansatz with one parameter reaches the FCI energy without noise. On `fake_brisbane` (Table 1) the witness rises with the idle delay, in step with the fraction of shots that lost an electron. The filter lowers the error $3.2\times$ below readout mitigation alone and $3.7\times$ below classical Hartree–Fock. Its gain grows with T1. Along the dissociation curve Hartree–Fock fails increasingly, while the filtered quantum estimate stays within 6–20 mHa, $11\times$ below Hartree–Fock at 2.5 Å.

Trapped-ion noise. On IonQ’s `aria-1` and `forte-1` noise models (2000 shots, 3 runs) the witness barely moves (+0.004, +0.005) and only 2–3% of the shots violate the electron number. Dropping them still lowers the error from ~ 30 to ~ 12 mHa, because number-violating outcomes sit at very high energy. The run-to-run spread is about ± 8 mHa.

Spin-resolved witnesses. H_2 also conserves $N_\uparrow$ and $N_\downarrow$ separately. Keeping one up and one down electron removes two more bit strings. Under depolarizing noise on the CXs the spin leak is 1.5–5%, and the second filter removes a further 34–37% of the error ($23.2 \rightarrow 14.7$ mHa at $p = 0.03$). On `fake_brisbane` the leak is below 1% and the gain is within the seed spread ($6.5 \rightarrow 5.9$ mHa). The residual ~ 5 mHa of Table 1 is in the $XXYY$ terms and in errors that preserve both numbers, not in spin leak.

Table 2: H_2 , error vs FCI (mHa), mean $\pm$ std over 10 seeds. Controlled noise acts on every CX.

noise	$\overline{qg}_Z$	kept	readout-mitigated	+ qg filter	+ ZNE	+ ZNE + qg
T1, $p = 0.03$	+0.029	0.94	36.6	2.0 ± 1.6	-4.0 ± 3.1	-2.0 ± 2.9
T1, $p = 0.10$	+0.099	0.81	128.9	7.4 ± 1.5	-5.7 ± 5.3	-0.6 ± 3.5
dephasing, $p = 0.10$	0.000	1.00	10.2	10.2 ± 1.7	2.1 ± 3.3	2.1 ± 3.3
depolarizing, $p = 0.03$	+0.007	0.96	60.8	23.2 ± 2.3	0.3 ± 4.5	-2.3 ± 3.8
depolarizing, $p = 0.10$	+0.021	0.86	193.5	81.3 ± 3.8	17.5 ± 8.2	-3.3 ± 8.8
readout, $p = 0.03$	0.000	0.89	-0.6	-0.6 ± 1.2	-1.9 ± 2.9	-1.9 ± 2.1
fake_brisbane	+0.011	0.90	20.6	6.5 ± 1.6	6.8 ± 5.9	2.0 ± 3.9
fake_brisbane, 50 μ s idle	+0.122	0.72	189.3	31.1 ± 1.3	176.6 ± 4.1	27.2 ± 4.1

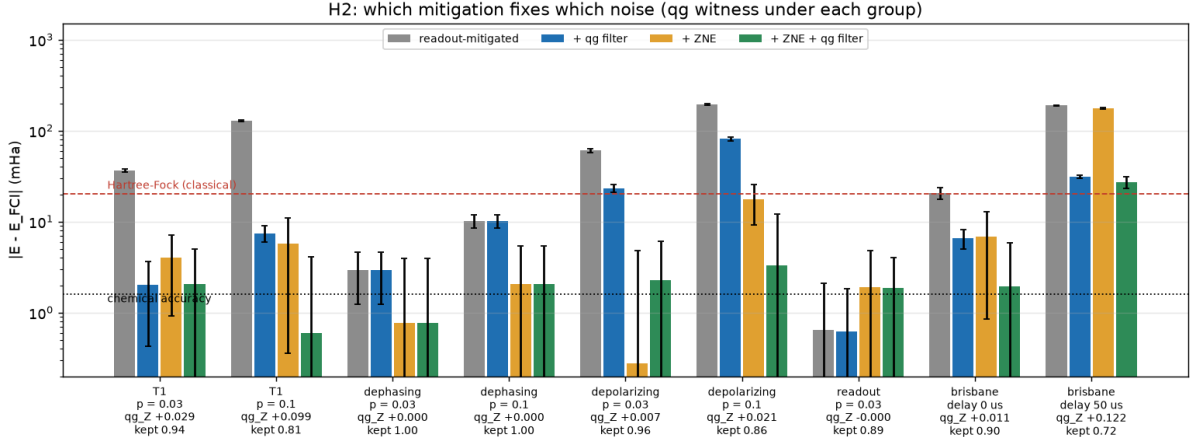


Figure 1: Energy error of H_2 by noise type and mitigation (log scale). Under each group, the qg witness and the kept fraction read from the unmitigated shots.

Where it stops. For LiH at 3.0 Å (6 qubits, ideal $\overline{qg}_Z = 1/3$) the ansatz needs 60 ECR gates. Every quantum estimate is then $\sim 9\times$ worse than Hartree-Fock, and the filter does not help. The witness explains why: $\overline{qg}_Z$ falls from +0.333 to +0.18, towards 0 and not towards +1. The dominant error is unital scrambling, which leaves the kept shots scrambled too.

5 Which noise does the filter fix?

Table 2 compares the filter with ZNE on the same H_2 state. Three regularities appear (Fig. 1).

- **The witness predicts whether the filter works.** Dephasing preserves the electron number. The witness stays at 0, every shot is kept, and the filter changes nothing; only ZNE helps. T1 moves $\overline{qg}_Z$ up by about p , and the filter removes 94–95% of the error with no extra circuits and less spread than ZNE.
- **The methods are complementary.** ZNE overshoots under T1, by -4 to -6 mHa, because amplitude damping is not linear in the fold count, and it doubles or triples the spread. Under strong depolarizing noise only the combination comes within a few mHa.
- **Idle decoherence defeats ZNE.** Gate folding does not scale a 50 μ s idle period (189 $\rightarrow$ 177 mHa). The witness flags it (+0.122), and the filter removes 84% of the error.

This gives a decision rule. If $\overline{qg}_Z$ leaves $1 - 2N/n$ or the kept fraction drops, apply the filter, which is free. If the error stays large with the kept fraction near 1, the remaining noise preserves the symmetry and needs ZNE.

Table 3: 1D Hubbard on `fake_brisbane`: absolute error of D and I vs the noiseless Trotter circuit (5 seeds).

	D , 1 step	D , 2 steps	I , 1 step	I , 2 steps
readout-mitigated	.081	.056	.206	.111
+ N filter	.050	.029	.124	.070
+ spin filters	.039	.016	.100	.047
+ ZNE	.031	.031	.036	.010
+ ZNE + spin filters	.009	.006	.035	.071

6 Materials: 1D Hubbard dynamics

The Fermi–Hubbard chain with $L = 4$ sites ($J = 1$, $U = 2$, 8 qubits, spin blocks in Jordan–Wigner) evolves by first-order Trotter steps ($dt = 0.25$) from a charge-density wave with sites 0 and 2 doubly occupied, a standard hardware benchmark [8]. The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$, and here *every* observable of interest (charge imbalance I , double occupancy D) is diagonal in Z , so the filters act on all of them. The reference is the noiseless output of the same Trotter circuit.

On the all-to-all device (20 CX per step) the filters roughly halve the error. The spin filters add a further 30% at depth, as the spin leak grows from 0 to 8%. ZNE + spin filters gives the best double occupancy: time-averaged error 0.003 vs 0.019 raw. The witnesses stay at 0 because the noise is unital, but the kept fraction falls from 0.88 to 0.50.

On `fake_brisbane` routing the ladder of hops and interactions costs 56 ECR per step, and routing creates spin leak: 8, 16, 41, 45 and 47% of the right- N shots sit in the wrong spin sector after 1, 2, 4, 6 and 8 steps (H_2 : < 1%). The spin filters therefore beat the N filter clearly (Table 3). From 4 steps (≥ 188 ECR, 18% of shots kept) every method sits on the noise floor. $D \rightarrow 1/4$, the value of the maximally mixed state inside the spin sector, and $I \rightarrow 0$ (Fig. 2). With the current `fake_brisbane` snapshot mostly the spin-down register drifts towards +1; with an older snapshot both do. The per-register witnesses report the calibration, not a property of the method.

7 Constrained optimization: QAOA with “exactly K ”

Maximum K -vertex cover asks for exactly K of n vertices that touch the most edges. With $x_i = (1 - Z_i)/2$ the constraint is $\overline{qg}_Z = 1 - 2K/n$ (0.25 for $n = 8$, $K = 3$). We compare standard penalty QAOA [9]; a *budget start*, with every qubit at $qg_Z = 1 - 2K/n$ and the mixer rotated about that axis, which needs no classical pre-solve; a warm start with $qg_Z^{(i)} = 1 - 2c_i$ from the LP relaxation c [10], clipped to $|qg_Z| \leq 0.5$ (Egger’s $\varepsilon = 0.25$); and the constraint-preserving XY-mixer ansatz [11] started from a Dicke state [12]. Angles are optimized without noise, and 4000 shots are taken on 5 random graphs $G(8, 1/2)$.

On the all-to-all device (Fig. 3) the filter doubles the probability that a shot is optimal for the XY ansatz: $0.150 \rightarrow 0.278$ at depth 1 and $0.177 \rightarrow 0.366$ at depth 2. Because that ansatz is 100% feasible without noise, its kept fraction (0.54) is a pure error witness. Penalty QAOA at depth ≤ 2 does no better than a random feasible guess after filtering (probability of the optimum 0.10–0.18 vs 0.096, approximation ratio 0.78 vs 0.77). The budget start raises the raw feasibility but not the quality. The warm start is the best penalty variant, but its LP relaxation is already integral on 3 of 5 graphs, so its value is classical. On `fake_brisbane` (134–497 ECR) every variant ends at the random-feasible level. The kept fraction (0.24–0.33) is then close to $\binom{8}{3}/2^8 = 0.22$, which tells the user that no signal is left. A greedy algorithm reaches 98.5% of the optimum, and brute force is instant.

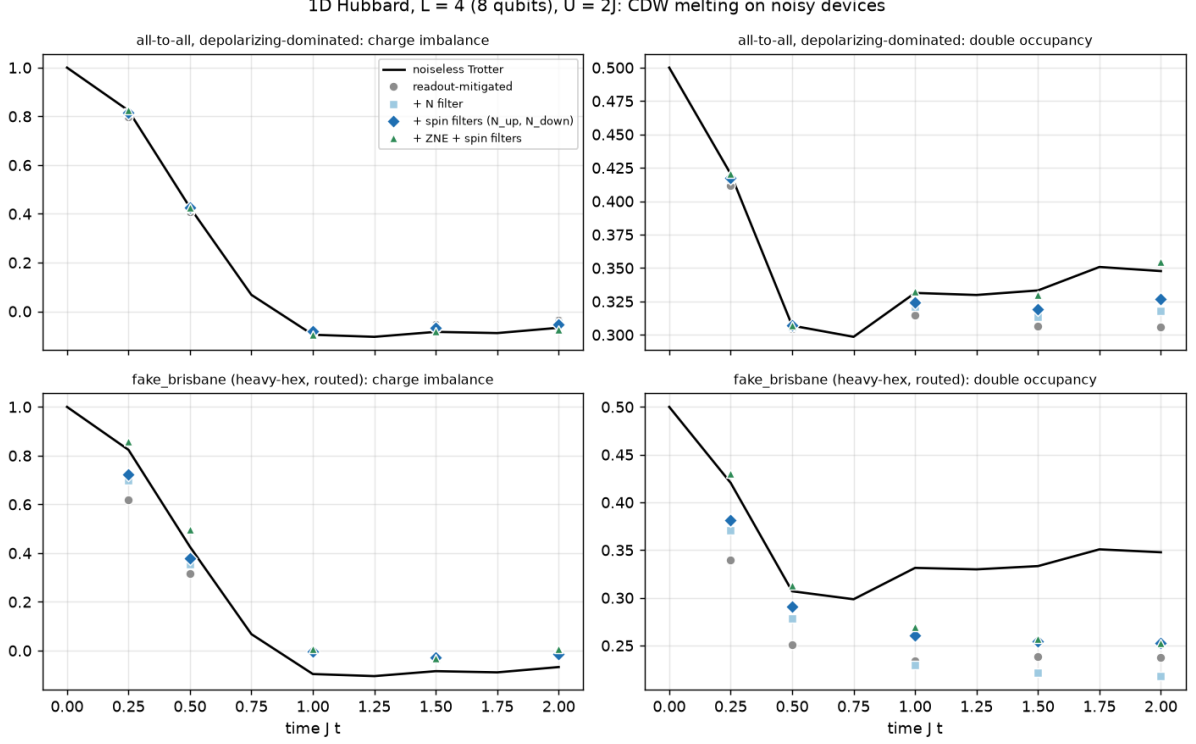


Figure 2: Charge-density-wave melting in the 1D Hubbard model on two simulated devices. Black: noiseless Trotter circuit.

8 Calibrating what the witness relies on

Eq. (2) makes the witness depend on a and b . A real qubit also idles with an excited population p , with $qg_Z^{\text{eq}} = 1 - 2p = \tanh(hf/2k_B T_{\text{eff}})$ [13]. The standard calibration prepares “|0>” (the thermal state) and “|1>” (X on it). From Z-basis data alone it sees only $a + bqg_Z^{\text{eq}}$ and bqg_Z^{eq} , so it counts thermal population as readout error.

Heralded sweep. Measure once (herald), apply I, X or $R_y(\pi/2)$, wait t , rotate back and measure again. For outcomes $r_1, r_2 = \pm 1$ at $t = 0$ and a QND herald,

$$m = \mathbb{E}[r_1] = a + bq, \quad V = \mathbb{E}[r_1 r_2 | I] - m^2 = b^2(1 - q^2), \quad C_X = \mathbb{E}[r_1 r_2 | X] = a^2 - b^2, \quad (3)$$

with $q = qg_Z^{\text{eq}}$. Hence $u = bq = (m^2 - V - C_X)/(2m)$, $b = \sqrt{V + u^2}$ and $a = m - u$. The delays add T_1 (after X) and T_2 (the decay of qg_X after $R_y(\pi/2)$). A joint maximum-likelihood fit of all joint counts returns (a, b, q, T_1, T_2) with the same circuit and shot budget as the standard suite. The model is checked against a Qiskit Aer circuit that prepares the thermal state by purification.

Result. With $T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$, $e_{01} = 0.015$, $e_{10} = 0.04$ and a 5 GHz qubit (Fig. 4), the standard suite reports $T_{\text{eff}} = 0$ at every temperature and overestimates e_{01} by exactly pb (0.0915 at $p = 0.08$). The heralded sweep is unbiased: e_{01} is 0.0148 – 0.0152 and T_{eff} is 52.3 ± 1.0 , 69.0 ± 0.8 and 98.2 ± 0.9 mK at true 52, 69 and 98 mK. Both protocols give unbiased T_1 and T_2 ; the joint fit is about $1.5\times$ more precise. If the herald is not QND, with 5% relaxation of |1> during it, the estimates move by about 0.001. Separating thermal population by repeated measurement is known practice. What qg adds is the closed form and T_{eff} read directly from qg_Z^{eq} . The protocol needs mid-circuit measurement.

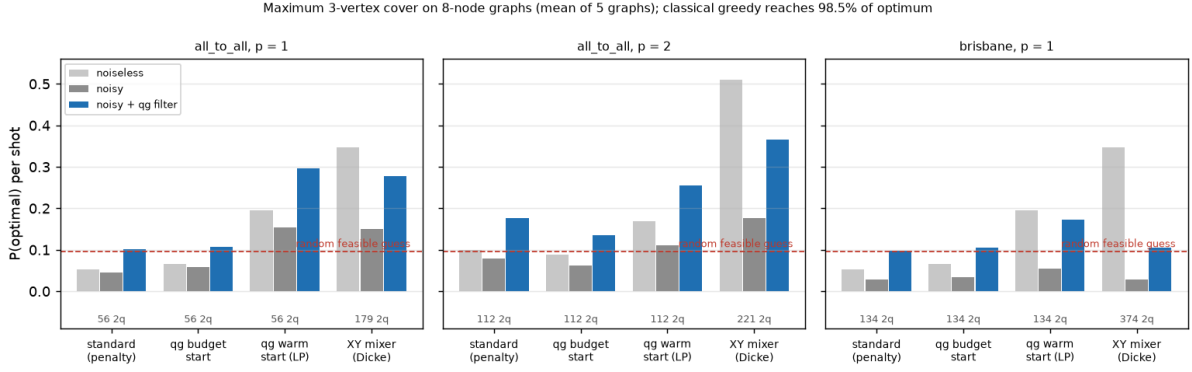


Figure 3: Probability that one shot is an optimal cover, noiseless, noisy, and noisy + qg filter. The dashed line is a random feasible guess. The number of two-qubit gates is under each group.

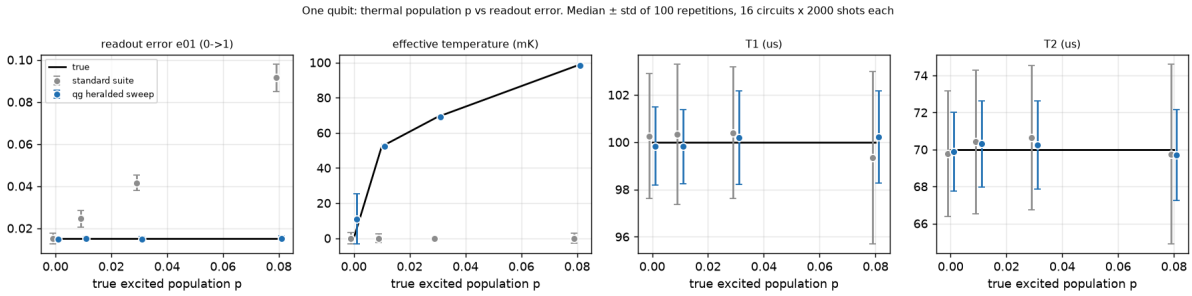


Figure 4: Standard calibration (gray) vs heralded qg sweep (blue); black: truth. Median $\pm$ std of 100 repetitions, 16 circuits $\times$ 2000 shots each.

9 Trapped-ion noise models

IonQ devices are all-to-all, so the routing that dominated Sections 6 and 7 disappears. We reran the Hubbard, QAOA and H_2 experiments on IonQ’s cloud simulator with its **aria-1** and **forte-1** noise models (2000 shots per circuit, the simulator’s limit; vendor noise models, not hardware).

Hubbard. Without routing the spin leak is small at shallow depth (0.1% at one step, against 8% on **fake_brisbane**), as expected. It still grows with depth (8.3% and 11.5% at six steps), and there the spin filters beat the N filter. Averaged over 1, 2, 4 and 6 steps, the error of the double occupancy is 0.022 / 0.016 / 0.013 (readout-mitigated / N filter / spin filters) on **aria-1** and 0.029 / 0.021 / 0.017 on **forte-1**. The register witnesses stay at 0 (unital noise); the kept fraction does the reporting.

QAOA. The filter raises the probability that a shot is optimal for the XY ansatz from 0.155 to 0.276 (depth 1) and from 0.174 to 0.363 (depth 2) on **aria-1**, and from 0.120 to 0.256 and 0.111 to 0.293 on **forte-1**. This reproduces the all-to-all result of Section 7, while on **fake_brisbane** the same circuits fell to the random-guess level: connectivity is what keeps the signal.

H_2 : filter vs ZNE at a small shot budget. Folding CX gates does not amplify the noise on IonQ: the service recompiles circuits written in the standard gate set and the folded pairs vanish (a Bell pair keeps $P(00) + P(11) = 0.99$ with 1 to 21 CX). We therefore fold the native Mølmer-Sørensen gates, $MS \rightarrow MS MS(\phi_0 + \frac{1}{2}, \phi_1) MS$, which the service runs as written (raw error 31 $\rightarrow$ 84 $\rightarrow$ 133 mHa at scales 1, 3, 5). Over six runs per model (Table 4) the filter halves the error in every run and its mean is below Hartree-Fock, whereas ZNE is nearly unbiased but

Table 4: H_2 on IonQ noise models, error vs FCI (mHa), mean $\pm$ std over 6 runs, 2000 shots per circuit. Hartree-Fock: 20.3 mHa.

noise model	readout-mitigated	+ qg filter	+ ZNE (MS folding)	+ ZNE + qg
aria-1	31.5 ± 7.8	14.7 ± 5.8	4.2 ± 20.3	5.5 ± 16.9
forte-1	34.8 ± 6.3	13.9 ± 5.4	6.6 ± 19.0	3.0 ± 17.8

its extrapolation amplifies shot noise to ± 17 – 20 mHa, so its typical error is larger. This reverses Section 5, where 20,000 shots per circuit favoured ZNE + qg: the preferred method depends on the shot budget.

10 Beyond symmetries: syndromes as qg witnesses

In a code-capacity model with amplitude damping γ and dephasing p per round, the single-qubit witness is $qg_X = \sqrt{1-\gamma}(1-2p)$ for $|+\rangle$ and $qg_Z = -1 + 2\gamma$ for $|1\rangle$. Every stabilizer expectation of the codes we consider is a power of it: $\langle X_1 X_2 \rangle = qg_X^2$ for the phase-flip code, and $\langle XXXX \rangle = qg_X^4$, $\langle Z_0 Z_1 \rangle = 1 - 2\gamma(1-\gamma)$ for the 4-qubit amplitude-damping code of Leung *et al.* So the qg of each syndrome ancilla inverts in one line to (γ, p) while the code runs.

Which code. Repetition codes do not correct T1; the Leung code does, with logical infidelity $\approx 0.92\gamma^2$ instead of $\gamma/3$, but it is fragile to dephasing and beats no code only for $p < \gamma/4$. For one idle round ($\gamma = t/T_1$, $p = t/2T_\phi$) this is a rule on T_2/T_1 alone: Leung code if $T_2 > T_1$, no code if $0.4T_1 < T_2 < T_1$, phase-flip code if $T_2 < 0.4T_1$. A transmon with $T_1 = 100 \mu\text{s}$ and $T_2 = 70 \mu\text{s}$ is in the “no code” band; trapped ions (no T1) never need the Leung code.

Tracking drift. When T_2/T_1 switches between the three regimes (telegraph model, 400 windows of 1000 rounds), choosing the code from the running syndromes costs 13% more infidelity than an oracle with essentially no extra shots, against 20% for recalibrating every 10 windows and 58% for calibrating once. A hybrid that probes only while no code is running reaches 7%, within 0.6 points of probing every window with $4\times$ fewer probe shots. These are idealized models (perfect syndrome extraction, synthetic drift), and the codes are known; the contribution is the qg reading and the rules.

11 Discussion and limitations

1. *No quantum advantage.* FCI, exact diagonalization, brute force and greedy search are exact or near-exact at every size studied. The comparisons rank strategies for a quantum computation.
2. *The filter is known* [2, 3]. The contributions are the witness as a decision rule, the systematic comparison with ZNE and with classical baselines, and the calibration protocol.
3. *Walls.* Beyond ~ 60 (LiH) or ~ 190 (Hubbard, heavy-hex) noisy two-qubit gates, unital scrambling dominates and no method helps. The witness then reports a drift towards 0 and a kept fraction near $\binom{n}{N}/2^n$.
4. *Rotated-basis terms* such as $XXYY$ cannot be filtered with Z-basis shots; they hold most of the residual H_2 error.
5. *Simulation only.* All devices are calibration-based or vendor noise models, including the IonQ runs, which used IonQ’s own simulator. Hardware runs are prepared (IBM, IonQ)

but not yet performed. The characterization protocol assumes perfect gates and a nearly QND herald.

6. *Shot budget.* Whether the filter or ZNE is preferable depends on the number of shots: at 2000 shots per circuit ZNE’s variance dominates, at 20,000 its lower bias wins.
7. *Snapshot dependence.* Per-register asymmetries depend on the calibration snapshot of the fake backend.

12 Code availability

All scripts, data and the regression tests that pin every number (859 tests at the time of writing) are at <https://github.com/Viny2030/qang> (MIT license): `examples/chemistry_*.py`, `error_mitigation_qg_vs_zne.py`, `hubbard_trotter_qg_filters.py`, `qaoa_k_constraint_qg.py`, `hardware_characterization_qg.py`, `qec_leung_code_t1_qg.py`, `qec_syndrome_drift_tracking_qg.py`, `ionq_validation.py`, `ionq_sim_hubbard_qaoa.py` and `ionq_sim_zne_grover.py`; the IonQ simulator results are recorded in `examples/data/ionq_sim_results.json`. Sections 20, 21, 24, 28–33, 38 and 39 of `RESEARCH_NOTES.md` give further detail, and `notebooks/qang_avances_colab.ipynb` reproduces fast versions in Colab.

Acknowledgments and use of AI tools

Generative AI tools (Anthropic Claude Opus 5.5) assisted with code generation, numerical experiments, test expansion and drafting of this text. The author framed the research questions, made the design decisions, and reviewed, edited and validated all AI-assisted output, including every number reported here.

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